



Predicting Outcomes in Professional League of Legends Matches

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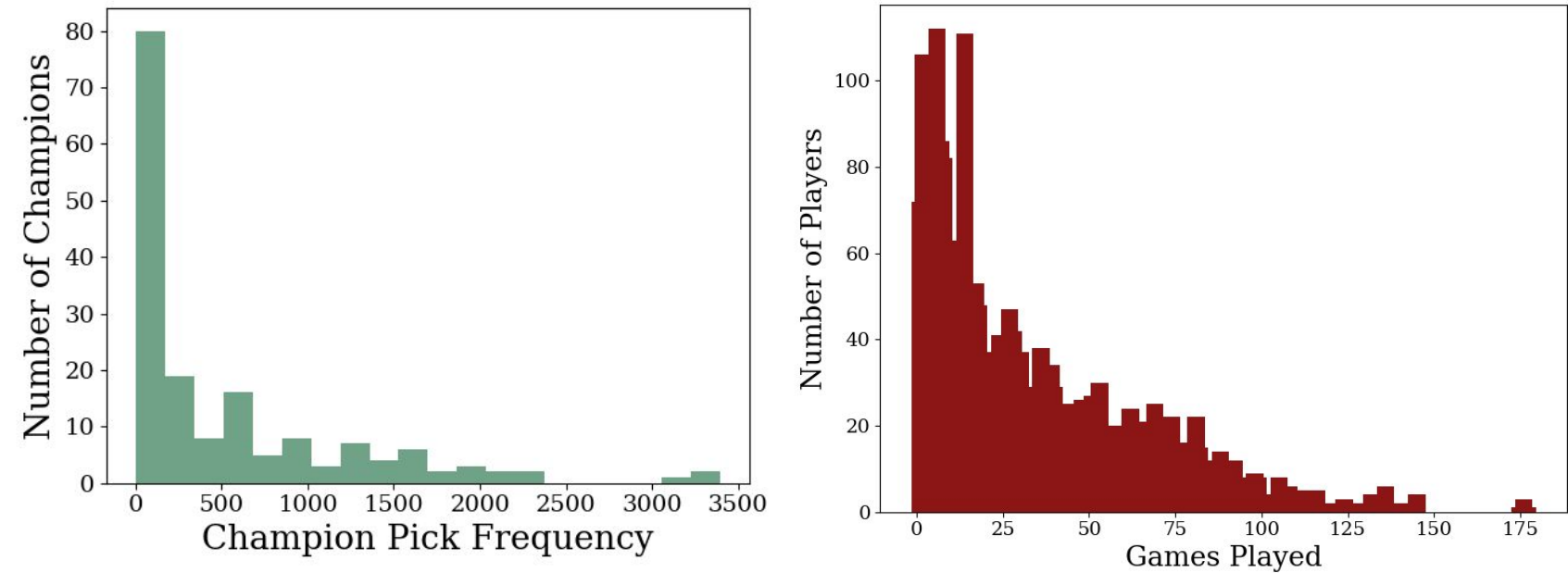
Project Overview

League of Legends, the world’s largest esport, boasts over 150 million registered players and a vibrant professional scene with millions of concurrent viewers. Each professional match provides valuable data, particularly the pre-match drafting phase, where teams select their champions. Our research focuses on predicting match outcomes based solely on pre-match data, using machine learning models to optimize prediction accuracy and identify impactful features of successful drafts.

Previous studies on League of Legends win prediction have explored both pre-match and in-game data, with varying results. Do et al. (2021) utilized pre-match data in ranked games, achieving 75.1% accuracy, but did not examine professional matches. Ani et al. (2019) incorporated in-game data and pre-match features, achieving a combined accuracy of 99.75% for professional matches. Costa et al. (2021) found that player statistics were more predictive than bans or champion picks, reporting an AUC of 0.97. Our work improves upon these findings by exclusively using professional match data and achieving higher classification accuracy.

Datasets & Features

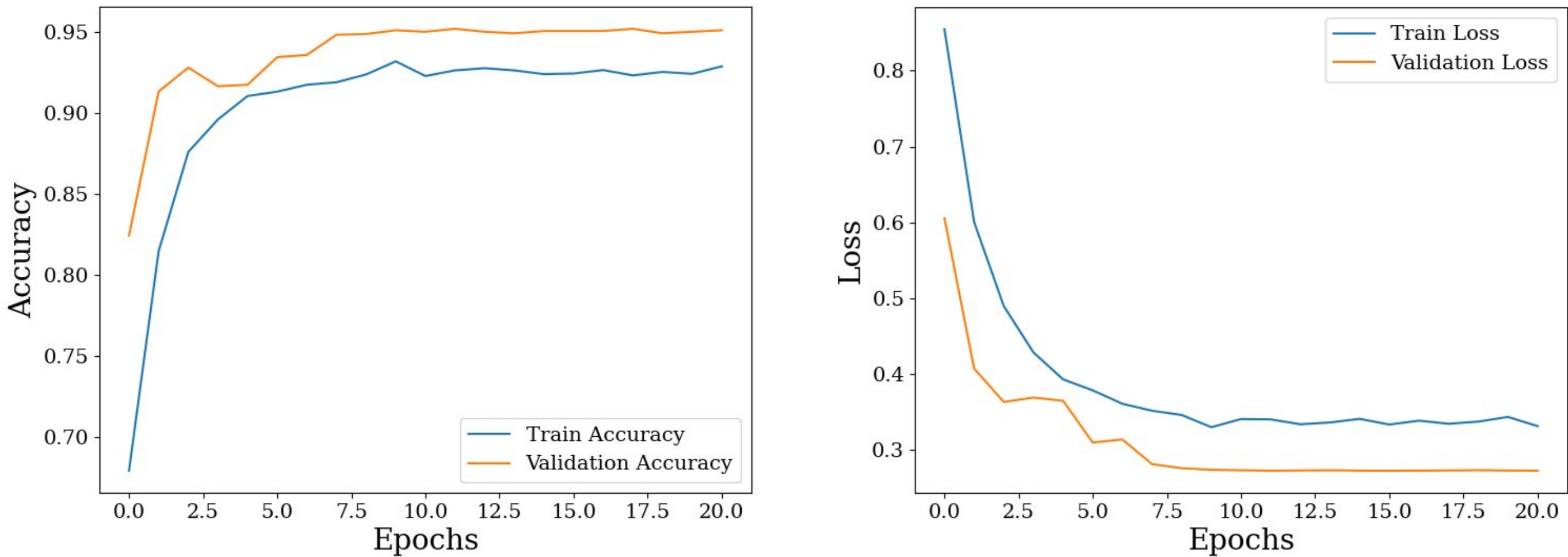
Our dataset, sourced from Oracle’s Elixir, contains data from 8,562 professional matches played in 2024, primarily focusing on major leagues. After pruning for completeness, we analyzed data from 2,652 unique players and 528 teams. We explored various input vector formats, including one-hot encoding of champion selections, win rates, and advanced embedding techniques using Word2Vec. These approaches allowed us to represent team compositions and player statistics in a compact, information-rich manner.



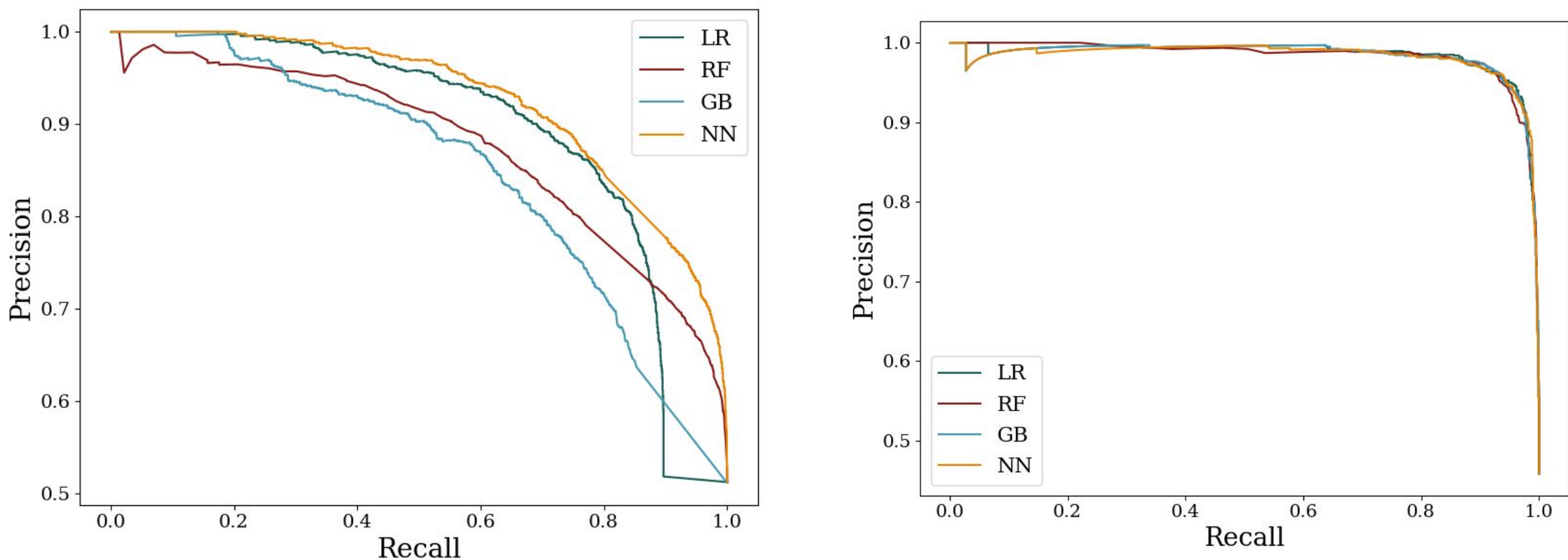
Methods & Metrics

Our analysis employed a diverse set of machine learning models trained on pre-match data, focusing on classification accuracy and other performance metrics. Our models included logistic regression, random forest, gradient boosting, and a neural network. Logistic regression was used for its simplicity and interpretability, while random forest and gradient boosting leveraged ensemble learning to enhance performance. The neural network model incorporated layers with dropout, ReLU activation, and batch normalization to prevent overfitting.

To evaluate model performance, we used a randomized 80/20 train-test split. We measured model accuracy, area under the receiver operating characteristic curve (AUC), precision, recall, and F1 score. These metrics provided a comprehensive assessment of model effectiveness, robustness, and predictive power.



NN Train/Validation Accuracy & Loss on Match Vector + EMB



Precision-Recall for One-hot Win Rate and Match Vector + EMB

Results

Across all models, classification accuracy exceeded 95%, with random forest and the neural network achieving the best results. Random forest reached 96% accuracy with match vectors and embeddings, while the neural network demonstrated the highest AUC at 99%. Embeddings provided marginal benefits, suggesting that win rates and champion statistics sufficiently capture the predictive information. Precision, recall, and F1 scores were consistently strong across methods, validating the robustness of our approach.

Accuracy / AUC				
	LR	GB	RF	NN
Baseline (One-hot)	0.53 / 0.53	0.52 / 0.52	0.51 / 0.51	0.54 / 0.55
One-hot Win Rate	0.82 / 0.82	0.75 / 0.76	0.78 / 0.78	0.82 / 0.92
Team Vector	0.91 / 0.91	0.91 / 0.91	0.91 / 0.91	0.91 / 0.98
Match Vector	0.95 / 0.95	0.95 / 0.95	0.95 / 0.95	0.95 / 0.99
Match Vector + EMB	0.95 / 0.95	0.95 / 0.95	0.96 / 0.95	0.95 / 0.99

Discussion & Future Research

Most of our models had high accuracy, but the models that used the Match vector input format were the most successful. This seems to be due to our inclusion of the champion-specific player win rate, which we believe to be the most significant feature in predicting success. Other features such as each player’s seasonal win rate and each champion’s seasonal win rate are also important, which we also applied to the banned champions. Even though we initially believed that it would be important to consider synergies between pairs of champions, it seems that directly adding features for each champion is not significantly useful. Rather, we believe champion and player win rates encapsulate these synergies for us, since professional teams draft champions with compatibility in mind.

In regards to future work, we would like to find some way to utilize the datasets from previous years, which could increase the accuracy of the model, despite the game and champions being updated every season. Furthermore, we believe that cleaning the data by removing entries corresponding to less prevalent teams and players could also improve our model.